MD. AWLAD HOSSAIN

Lecturer in Mathematics • Applied Mathematics • Scientific Machine Learning

Physics-Informed Learning • Neural Operators • Learning-Based Control • Embedded Systems and Robotics

Dhaka, Bangladesh | mdawlad.du@gmail.com | +880 1776 041967

Google Scholar | ORCID: 0009-0008-6017-0948 | ResearchGate | GitHub | LinkedIn | Portfolio

RESEARCH PROFILE

Research in scientific machine learning and computational mathematics, focusing on physics-informed and operator-learning methods for the modelling, control and inverse analysis of nonlinear systems governed by partial differential equations. Interests include data-efficient surrogate modelling, reinforcement learning for constrained control and uncertainty-aware parameter estimation, with applications in environmental, thermal-fluid, energy and biomedical systems, and the deployment of learned models on embedded and robotic platforms. Experienced in university teaching of mathematics and statistics.

RESEARCH INTERESTS

- **Scientific machine learning:** PINNs, DeepONet and Fourier neural operators, hybrid numerical–learning methods
- **Learning-based control:** PDE-constrained control, deep reinforcement learning, control barrier functions
- **Inverse problems:** parameter estimation, identifiability, uncertainty quantification, sparse-data inference
- **Embedded systems and robotics:** on-device deployment of scientific machine learning models for control and biomedical screening, vision-based teleoperation
- **Computational modelling:** CFD and heat transfer, atmospheric and environmental simulation

EDUCATION

M.Sc. in Applied Mathematics **2023–2025**
University of Dhaka CGPA: 3.72/4.00

- **Thesis:** *Patient-Specific Modeling of Breast Cancer and Inverse Estimation of Tumor Thermophysical Properties Using Deep Learning*
- Estimated tumour thermophysical properties from patient-specific models with deep learning, framing diagnosis as an inverse heat-transfer problem.
- **Supervisor:** Dr. S. M. Chapal Hossain, Associate Professor, Department of Applied Mathematics

B.Sc. (Honours) in Applied Mathematics **2019–2023**
University of Dhaka CGPA: 3.51/4.00

- **Project:** *Recent Trend Analysis of Rainfall and Temperature over Sylhet Region Using Statistical Methods* (project record)
- **Supervisor:** Dr. Md. Abdus Samad, Professor, Department of Applied Mathematics

Secondary: HSC (2018), Kabi Nazrul Government College, GPA 5.00/5.00 • SSC (2016), Shantibag High School, GPA 5.00/5.00

ACADEMIC APPOINTMENTS

Lecturer in Mathematics **Jul 2026–Present**
Department of General Education, Canadian University of Bangladesh Dhaka, Bangladesh

- Teach undergraduate mathematics to engineering and business students; write LaTeX lecture notes built on applied problems.

Adjunct Faculty **May 2026–Sep 2026**
Uttara University Dhaka, Bangladesh

- Taught undergraduate statistics and mathematics; prepared outcome-based (OBE) examinations, tables of specification and marking schemes.

Research Assistant **Mar 2025–Present**
Institute for Advanced Research, United International University Dhaka, Bangladesh
Project: *Deep Reinforcement Operator Learning for Dynamic Control* | **PI:** Dr. Mahtab Uddin

- Developed a physics-informed dual-critic PPO controller with a DeepONet surrogate for river-pollution control [J1]: against standard PPO it cut integrated pollutant exceedance by 11.65% and violation duration by 10.85%,

with a surrogate relative L_2 error of 2.21%.

- Extended the approach to safe control with Fourier neural operators and control barrier functions for battery thermal-runaway cooling [U3].
- Contributed to DeepONet-guided RL stabilisation of the chaotic Kuramoto–Sivashinsky equation [J2].

Mathematics Instructor **Jan 2025–Dec 2025**
Mojaru – Creative Education Platform Bangladesh

- Designed interactive mathematics tutorials and mentored high-school students in analytical problem solving.

Resource Member **Jan 2023–Feb 2025**
National Oceanographic and Maritime Institute (NOAMI) Bangladesh

- Contributed to research and workshops on WRF weather prediction, deep learning and climate applications.

PUBLICATIONS

3 journal articles (1 first-author) • 1 conference paper • 3 manuscripts under review | † corresponding author

Journal Articles

[J1] **Hossain, M. A.**, Ahmed, N., Uddin, M., Eme, K. A., & Azhari, H. A. (2026). Physics-informed operator learning with dual-critic proximal policy optimization for time-sensitive PDE-constrained control: Application to river pollution management. *Results in Engineering*, 32, 113707. doi:10.1016/j.rineng.2026.113707 (open access)

[J2] Ahmed, N., Babu, M. A., Hossain, M. S., Fayz-Al-Asad, M., **Hossain, M. A.**, Ahmmed, M. M., Rahman, M. M., & Mahmud, M. (2026). Reinforcement Operator Learning (ROL): A hybrid DeepONet-guided reinforcement learning framework for stabilizing the Kuramoto–Sivashinsky equation. *PLOS ONE*, 21(1), e0341161. doi:10.1371/journal.pone.0341161

[J3] Ahmed, N., **Hossain, M. A.**, Babu, M. A., Fayz-Al-Asad, M., & Ahmmed, M. M. (2026). A physics-informed piecewise continuity network for solving the non-linear jerk equation: PiPC-Net. *Physica A: Statistical Mechanics and its Applications*, 681, 131117. doi:10.1016/j.physa.2025.131117

Conference Proceedings

[C1] **Hossain, M. A.**†, Ahmed, N., Babu, M. A., Ahmmed, M. M., Rahman, M. M., & Mahmud, M. (2026). Piecewise neural network model to obtain large interval solution to the Lorenz system. In *Proc. 6th Int. Conf. on Trends in Computational and Cognitive Engineering (TCCE 2024)*, Lecture Notes in Networks and Systems, vol. 1588, pp. 277–293. Springer, Singapore. doi:10.1007/978-981-95-1069-6_20

Manuscripts Under Review

[U1] **Hossain, M. A.**, Uddin, M., Hossain, M. S., Mostafa, G., & Imran, M. M. H. MorphCal: Calibrated multiview electrocardiogram abnormality screening with embedded system evaluation. *Computers & Electrical Engineering*.

[U2] **Hossain, M. A.**, Adnan, M. I., Hossain, M. S., & Uddin, M. Identifiability-aware uncertainty-weighted physics-informed neural networks for sparse inverse parameter estimation in advection–diffusion–reaction systems. *Journal of Applied Mathematics and Computing*.

[U3] **Hossain, M. A.**, Ahmed, N., Uddin, M., Eme, K. A., & Azhari, H. A. FNO-CBF-RL for battery thermal runaway: Safe neural-operator reinforcement learning for energy-efficient cooling control. *Applied Mathematics and Computation*.

TEACHING

Courses Taught

- **Canadian University of Bangladesh:** several undergraduate mathematics courses each semester for engineering and business programmes, covering linear algebra and vector analysis, calculus and analytic geometry, and foundation mathematics.
- **Uttara University:** MAT 0541 1103 Mathematics II (coordinate geometry, ordinary differential equations, integration); GED-05422109 Statistics.

Curriculum and Assessment

- Write lecture notes, worksheets and worked-solution manuals in LaTeX; design OBE-aligned examinations with tables of specification and marking schemes.
- Use Python, MATLAB and Manim visualisations to connect mathematical concepts to computation and applications.

SELECTED RESEARCH PROJECTS

- **Vision-based robotic teleoperation (ongoing):** Designed and assembled a 6-DOF aluminium manipulator (high-torque servos, ESP32 controller, PCA9685 PWM driver over I^2C); developing stereo-camera hand tracking on an RDK X5 edge-AI computer for learning-based teleoperation.
- **Embedded biomedical AI:** Evaluated a calibrated multiview ECG abnormality-screening model on embedded hardware for low-cost, on-device screening [U1].
- **Hybrid weather prediction:** Contributed to a WRF–deep learning lightning-prediction model for Bangladesh hotspots, funded by the Ministry of Science and Technology, Government of Bangladesh.
- **Atmospheric simulation:** Simulated tropical cyclones with WRF and analysed their dynamical parameters.
- **Numerical and reduced-order modelling:** Wrote a shallow-water-equation solver in Fortran; reconstructed flow fields with POD and DMD.
- **Multiphysics simulation:** COMSOL studies of flow past multiple cylinders, heat transfer, structural response and Darcy porous-media transport; SolidWorks motion study of a slider-crank mechanism.

TECHNICAL SKILLS

Programming: Python, MATLAB, R, Fortran, C/C++, Mathematica; NumPy, Pandas, NetCDF4, WRF-Python, Manim

Scientific ML: PyTorch, TensorFlow, Keras, scikit-learn; PINNs, DeepONet, FNO, PPO and actor–critic RL; U-Net and ResNet for vision

Numerical Methods: PDE discretisation, inverse problems, parameter estimation, POD/DMD reduced-order modelling

Simulation Software: COMSOL Multiphysics, WRF, MPAS, CROCO, SolidWorks

Embedded and Robotics: ESP32 / ESP32-S3 (C/C++, Arduino framework), I^2C peripherals, PWM servo control (PCA9685), RDK X5 edge-AI board, stereo vision

Tools: LaTeX, Git/GitHub, Linux (Ubuntu, Mint, Manjaro), Power BI, Microsoft Office

ACADEMIC SERVICE AND OUTREACH

- Organising committee member, 12th National Undergraduate Mathematics Olympiad (NUMO 2021)
- Organising committee member, Applied Mathematics Premier League (AMPL), University of Dhaka
- Creator, *Functional Value*, a mathematics education YouTube channel

REFEREES

Dr. Mahtab Uddin — *Research Principal Investigator*
Associate Professor, Institute of Natural Sciences, United International University
mahtab@ins.uiu.ac.bd

Dr. S. M. Chapal Hossain — *M.Sc. Thesis Supervisor*
Associate Professor, Department of Applied Mathematics, University of Dhaka
chapal@du.ac.bd

Dr. Md. Abdus Samad — *B.Sc. Project Supervisor*
Professor, Department of Applied Mathematics, University of Dhaka
samad@du.ac.bd